Multi-Threshold Pass-Count Preservation under Bounded Score Perturbations

Research Architecture Runtime
`operator/multi-threshold`

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Abstract

We study the multi-threshold pass-count $C_{\mathbf{T}}(x) = |\{i : x \geq T_i\}|$ for a finite threshold list $\mathbf{T}$ under $|x' - x| \leq \varepsilon$. The count is invariant whenever every coordinate is outside its half-open unstable band inherited from scalar thresholding. Sharpness follows by flipping any single unstable cut. Lean formalization is Mathlib $\mathbb{R}$ (`REAL_MATHLIB`) (`LEAN_FULL`; `REAL_MATHLIB`).

1 Introduction

Multi-threshold selection aggregates many AboveThreshold bits into a Nat count. It is the natural finite extension of scalar thresholding and a reduction target for ordered cut-point / bucket operators.

2 Operator definition

Definition 1. *For finite $x \in \mathbb{R}$ and $\mathbf{T} = (T_0, \ldots, T_{n-1})$,*

$$C_{\mathbf{T}}(x) = \sum_{i=0}^{n-1} \mathbf{1}\{x \geq T_i\}.$$

Equality passes on each coordinate.

3 Perturbation model

Admissible neighbors satisfy $|x' - x| \leq \varepsilon$ with $\varepsilon \geq 0$.

4 Instability

Coordinate i is unstable when $x \in [T_i - \varepsilon, T_i + \varepsilon)$. The multi-threshold count is unstable whenever any coordinate is.

5 Main theorems

Theorem 2 (Multi-threshold preservation). *If $|x' - x| \leq \varepsilon$ and for every i either $x \geq T_i + \varepsilon$ or $x < T_i - \varepsilon$, then $C_{\mathbf{T}}(x') = C_{\mathbf{T}}(x)$.*

Theorem 3 (Sharpness). *If some j satisfies $x \in [T_j - \varepsilon, T_j + \varepsilon)$, then there exists admissible x' with $C_{\mathbf{T}}(x') \neq C_{\mathbf{T}}(x)$ (equivalently: the singleton list $[T_j]$ flips).*

6 Proofs

Proof of Theorem 2. Induct on the threshold list. The empty list has count 0. For $T :: \mathbf{T}'$, the head bit is preserved by scalar threshold preservation; the tail count is preserved by the inductive hypothesis on the remaining stable coordinates. $\square$

Proof of Theorem 3. On the singleton $[T]$, push x by $\pm\varepsilon$ toward the cut. If $x \geq T$ then $x' = x - \varepsilon$ fails; if $x < T$ then $x' = x + \varepsilon$ passes. $\square$

7 Comparison with prior operators

Relative to Threshold: same equality convention and half-open unstable band; new selected object is a Nat pass-count; proof pattern is coordinatewise reduction. Relative to Argmax (reference): different geometry (cuts vs margin among alternatives).

8 Lean formalization summary

Module `Research.Operators.MultiThreshold.Preservation`: `multi_threshold_preservation`, `multi_threshold_sharpness`. Domain: Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

9 Certificate

`LEAN_FULL` at `lean/certificates/multi-threshold/multi-threshold-count-preservation/`.

10 Limitations

- Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).
- Unordered pass-count; ordered bucket indices are a separate operator.
- No DP / privacy claims.

References

References

[1] Repository: `implementation/src/operators/multi_threshold/`.